The Bombay Salesian Society's DON BOSCO INSTITUTE OF TECHNOLOGY
(An Autonomous Institute affiliated to University of Mumbai)
Department of Information Technology

Experiment No. 6 - Hypothesis Testing

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Aim

To formulate and test two hypotheses using a two-tailed Z-test on sample data across three topics, compute the Z-statistic and p-value for each hypothesis, and determine whether there is sufficient statistical evidence to reject or accept the null hypothesis at a significance level of $\alpha = 0.05$.

Theory

1. What is Hypothesis Testing? Hypothesis testing is a formal statistical procedure used to make decisions about a population based on sample data. It allows us to determine whether an observed difference between two groups is statistically significant or merely due to random chance.

The process involves formulating two competing hypotheses and using sample evidence to decide which one is supported.

2. Types of Hypotheses: Every hypothesis test involves two complementary statements:

- **Null Hypothesis (H_0)** — The default assumption that there is no significant difference between the two groups being compared. It assumes any observed difference is due to chance.
- **Alternative Hypothesis (H_1)** — The claim that there is a real, statistically significant difference between the two groups.

The goal of hypothesis testing is to find enough evidence in the sample data to reject H_0 in favor of H_1.

3. Control Group vs Experiment Group: In this experiment, each topic involves two groups of 16 subjects/observations each:

- **Control Group** — The baseline group that receives no special treatment or intervention.
- **Experiment Group** — The group that receives the treatment or is exposed to a different condition.

Group statistics (mean and standard deviation) are computed from the raw 16-value samples using standard formulas.

4. The Z-Test: A Z-test is used to compare the means of two groups when the sample size is known and the population standard deviation is available or can be estimated. It is appropriate here because each group has $n = 16$ observations and a pooled standard deviation is computed.

The steps are:

Step 1 — Compute the Standard Error (SE):

$SE = SD / \sqrt{n}$

Where SD is the pooled standard deviation of the sample and n is the sample size. SE measures how much the sample mean is expected to vary from the true population mean.

Step 2 — Compute the Z-Statistic:

$Z = |Mean(Experiment) - Mean(Control)| / SE$

The Z-statistic measures how many standard errors apart the two group means are. A higher Z value indicates a greater difference between the groups relative to the variability in the data.

Step 3 — Determine the p-value: The p-value is the probability of observing a Z-statistic as extreme as the one computed, assuming H_0 is true. In a two-tailed test, both directions (positive and negative) are considered.

The Z-statistic is mapped to an approximate p-value using a standard normal distribution lookup table.

5. Two-Tailed Test: A two-tailed test checks for differences in both directions — whether the experiment group mean is significantly higher or lower than the control group mean. This is more conservative than a one-tailed test and is used when the direction of difference is not assumed in advance.

The rejection region exists on both tails of the normal distribution.

6. Decision Rule — Significance Level $\alpha = 0.05$: The significance level α defines the threshold for rejecting H_0:

Condition	Decision
$p < 0.05$	Reject H_0 — statistically significant difference exists
$p < 0.02$	Strongly reject H_0 — very strong evidence against H_0
$p \geq 0.05$	Fail to reject H_0 — insufficient evidence to claim a difference

If $p < \alpha$, the result is considered statistically significant, meaning the observed difference is unlikely to have occurred by chance.

7. Interpretation Badge System: Results are classified into decision categories based on p-value thresholds:

- **Strong Rejection** — $p < 0.02$, very strong statistical evidence.
- **Rejection** — $0.02 \leq p < 0.05$, statistically significant evidence.
- **Fail to Reject** — $p \geq 0.05$, insufficient evidence.

8. Caveats and Assumptions:

- The SD used in SE computation is treated as the pooled standard deviation across both groups.
- The Z-statistic uses the absolute difference of means (no direction assumed), consistent with a two-tailed framework.
- p-values are approximated using a piecewise lookup table mapped to the standard normal distribution; exact results would require computing the precise normal CDF. Slight differences from exact p-values may exist but do not affect the overall decision in most cases.

Execution

The following steps were performed across two components — raw data analysis and hypothesis testing:

Component 1 — Raw Data and Group Statistics (sample_data_3_topics.html):

Step 1 — Dataset Structure: Raw sample data was prepared for three topics. For each topic, 16 control observations and 16 experiment observations were recorded. Each observation represented a subject or plot measurement relevant to that topic.

Step 2 — Group Statistics Computation: For each topic and each group (control and experiment), the following were computed from the 16 raw values:

- **Mean** — Average value of the group using the avg() function.
- **Standard Deviation** — Population standard deviation ($n = 16$) using the sd() function.

These statistics — (n, control mean, experiment mean, pooled SD) — serve as inputs for the hypothesis testing component.

Step 3 — Display: Per-subject/plot rows were displayed for both groups along with the computed group statistics, test type (Two-tailed Z-test), and unit of measurement for each topic.

Component 2 — Hypothesis Testing (hypothesis_testing_topics.html):

Step 4 — Input Summary Rows: Pre-aggregated summary rows (n, control mean, experiment mean, pooled SD) for each topic were loaded into the testing interface.

Step 5 — SE Computation: For each row, the Standard Error was computed as:

$$SE = SD / \sqrt{n}$$

Step 6 — Z-Statistic Computation: The Z-statistic was computed as:

$$Z = |\text{Mean(Experiment)} - \text{Mean(Control)}| / SE$$

Step 7 — p-value Mapping: The Z-statistic was mapped to an approximate two-tailed p-value using a standard normal distribution lookup table.

Step 8 — Decision: Each row was flagged based on the p-value:

- $p < 0.02$ → Strong Rejection of H_0
- $p < 0.05$ → Rejection of H_0
- $p \geq 0.05$ → Fail to Reject H_0

Results were displayed with interpretation badges and color-coded styling for clarity.

Hypotheses Tested:

Hypothesis 1:

- **H_0** — There is no significant difference in the mean values between the control group and the experiment group for Topic 1.
- **H_1** — There is a significant difference in the mean values between the control group and the experiment group for Topic 1.
- **Test:** Two-tailed Z-test, $\alpha = 0.05$
- **Decision:** *(Based on computed p-value — paste your result here)*

Hypothesis 2:

- **H_0** — There is no significant difference in the mean values between the control group and the experiment group for Topic 2.
- **H_1** — There is a significant difference in the mean values between the control group and the experiment group for Topic 2.
- **Test:** Two-tailed Z-test, $\alpha = 0.05$
- **Decision:** *(Based on computed p-value — paste your result here)*

Output

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127.0.0.1:5500/hypothesis_testing_topics.html

Hypothesis Testing Results Table For Multiple Topics

Select a topic to view its hypothesis test results (Two-tailed, α = 0.05)

Sleep & Memory | Fertilizer & Crop Yield | Noise-Cancelling & Focus

Control group
Avg recall (6h sleep)

Experiment group
Avg recall (8h sleep)

Test type
Two-tailed, α = 0.05

H_0: Sleep extension (8h vs 6h) does not improve memory recall scores
H_1: 8 hours of sleep improves memory recall compared to 6 hours

Sample size (n)	Control mean	Experiment mean	Pooled SD	Std error (SE)	Z-statistic	p-value (approx)	Decision (α=0.05)	Interpretation
4	18.2	20.1	5	2.50	0.76	0.37	Do not reject H_0	Difference likely due to randomness
9	18.4	21	4.8	1.60	1.63	0.07	Do not reject H_0	Weak evidence of improvement
16	18.3	21.5	4.5	1.13	2.84	0.0027	Reject H_0	Still insufficient evidence
25	18.5	22	4.2	0.84	4.17	0.0027	Reject H_0	Trending but not significant
36	18.4	22.3	4	0.67	5.85	0.0027	Reject H_0	Very close to significance
49	18.3	22.6	3.8	0.54	7.92	0.0027	Reject H_0	Statistically significant difference
64	18.5	22.9	3.5	0.44	10.06	0.0027	Reject H_0	Strong evidence of improvement
100	18.4	23.1	3.2	0.32	14.69	0.0027	Reject H_0	Very strong evidence

Do not reject H_0 No significant difference found   Reject H_0 Significant difference found   Reject H_0 Strong/very strong evidence

Sample Data For Three Hypothesis Testing Topics

Topic 1: Sleep & Memory | Topic 2: Fertilizer & Crop Yield | Topic 3: Headphones & Focus

H_0: There is no significant difference in memory recall between participants sleeping 6h vs 8h per night.
H_1: Participants sleeping 8h recall significantly more words than those sleeping 6h.

Total participants / plots
32 (16 per group)

Measurement unit
words recalled (out of 30)

Test type
Two-tailed Z-test

Significance level
α = 0.05

Control group — 6 hours sleep

Participants slept their usual 6h/night for 2 weeks, then took a 30-word recall test.

ID	Gender	Age	Words recalled
P01	Male	22	17
P02	Female	24	19
P03	Male	21	16
P04	Female	23	18
P05	Male	25	20
P06	Female	22	15
P07	Male	23	18
P08	Female	24	17
P09	Male	21	19
P10	Female	22	16
P11	Male	24	18
P12	Female	23	17
P13	Male	22	20
P14	Female	21	15
P15	Male	25	19
P16	Female	22	18
Mean			17.63
Std dev			1.54
n			16

Experiment group — 8 hours sleep

Participants extended sleep to 8h/night for 2 weeks, then took the same 30-word recall test.

ID	Gender	Age	Words recalled
P17	Male	22	23
P18	Female	24	25
P19	Male	21	22
P20	Female	23	24
P21	Male	25	26
P22	Female	22	21
P23	Male	23	24
P24	Female	24	23
P25	Male	21	25
P26	Female	22	22
P27	Male	24	24
P28	Female	23	23
P29	Male	22	26
P30	Female	21	21
P31	Male	25	25
P32	Female	22	24
Mean			23.63
Std dev			1.54
n			16

Each row is one participant / plot. Mean and SD are computed from the 16 data points shown and feed directly into the Z-test table.

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Hypothesis Testing Results Table For Multiple Topics

Select a topic to view its hypothesis test results (Two-tailed, $\alpha = 0.05$)

Sleep & Memory | **Fertilizer & Crop Yield** | Noise-Cancelling & Focus

Control group
Yield (conventional) kg/acre

Experiment group
Yield (organic) kg/acre

Test type
Two-tailed, $\alpha = 0.05$

H_0: Organic fertilizer does not increase rice yield compared to conventional fertilizer
H_1: Organic fertilizer significantly increases rice yield

Sample size (n)	Control mean	Experiment mean	Pooled SD	Std error (SE)	Z-statistic	p-value (approx)	Decision (α=0.05)	Interpretation
4	320	335	40	20.00	0.75	0.37	Do not reject H_0	Difference likely due to randomness
9	321	338	38	12.67	1.34	0.13	Do not reject H_0	Weak evidence of higher yield
16	320	340	36	9.00	2.22	0.016	Reject H_0	Still insufficient evidence
25	322	343	33	6.60	3.18	0.0027	Reject H_0	Trending but not significant
36	320	345	30	5.00	5.00	0.0027	Reject H_0	Very close to significance
49	321	347	28	4.00	6.50	0.0027	Reject H_0	Statistically significant difference
64	320	349	25	3.13	9.28	0.0027	Reject H_0	Strong evidence of higher yield
100	321	352	22	2.20	14.09	0.0027	Reject H_0	Very strong evidence

Do not reject H_0 No significant difference found | Reject H_0 Significant difference found | Reject H_0 Strong/very strong evidence

Sample Data For Three Hypothesis Testing Topics

Topic 1: Sleep & Memory | **Topic 2: Fertilizer & Crop Yield** | Topic 3: Headphones & Focus

H_0: Organic fertilizer does not produce a significantly different rice yield compared to conventional fertilizer.
H_1: Organic fertilizer produces a significantly higher rice yield than conventional fertilizer.

Total participants / plots	Measurement unit	Test type	Significance level
32 (16 per group)	**yield in kg/acre**	**Two-tailed Z-test**	**$\alpha = 0.05$**

Control group — conventional fertilizer

Plots treated with standard NPK chemical fertilizer. Yield measured after one full growth cycle.

ID	Soil	Location	Yield (kg/acre)
Plot 01	Clay	North	310
Plot 02	Loam	South	325
Plot 03	Sandy	East	308
Plot 04	Loam	West	320
Plot 05	Clay	North	315
Plot 06	Sandy	South	305
Plot 07	Loam	East	328
Plot 08	Clay	West	318
Plot 09	Sandy	North	312
Plot 10	Loam	South	322
Plot 11	Clay	East	316
Plot 12	Sandy	West	307
Plot 13	Loam	North	330
Plot 14	Clay	South	313
Plot 15	Sandy	East	309
Plot 16	Loam	West	324
Mean			316.38
Std dev			7.57
n			16

Experiment group — organic fertilizer

Plots treated with compost-based organic fertilizer at equivalent nutrient levels. Same growth cycle.

ID	Soil	Location	Yield (kg/acre)
Plot 17	Clay	North	338
Plot 18	Loam	South	352
Plot 19	Sandy	East	333
Plot 20	Loam	West	348
Plot 21	Clay	North	341
Plot 22	Sandy	South	330
Plot 23	Loam	East	355
Plot 24	Clay	West	343
Plot 25	Sandy	North	336
Plot 26	Loam	South	349
Plot 27	Clay	East	340
Plot 28	Sandy	West	331
Plot 29	Loam	North	358
Plot 30	Clay	South	337
Plot 31	Sandy	East	332
Plot 32	Loam	West	350
Mean			342.06
Std dev			8.67
n			16

Each row is one participant / plot. Mean and SD are computed from the 16 data points shown and feed directly into the Z-test table.

Hypothesis Testing Results Table For Multiple Topics

Select a topic to view its hypothesis test results (Two-tailed, α = 0.05)

Sleep & Memory | Fertilizer & Crop Yield | Noise-Cancelling & Focus

Control group
Completion time (no headphones) min

Experiment group
Completion time (headphones) min

Test type
Two-tailed, α = 0.05

H_0: Noise-cancelling headphones do not improve task completion time
H_1: Noise-cancelling headphones reduce task completion time

Sample size (n)	Control mean	Experiment mean	Pooled SD	Std error (SE)	Z-statistic	p-value (approx)	Decision (α=0.05)	Interpretation
4	42	39	8	4.00	0.75	0.37	Do not reject H_0	Difference likely due to randomness
9	42	38	7.5	2.50	1.60	0.07	Do not reject H_0	Weak evidence of faster completion
16	43	37	7	1.75	3.43	0.0027	Reject H_0	Still insufficient evidence
25	42	36	6.5	1.30	4.62	0.0027	Reject H_0	Trending but not significant
36	42	36	6	1.00	6.00	0.0027	Reject H_0	Very close to significance
49	43	35	5.5	0.79	10.18	0.0027	Reject H_0	Statistically significant difference
64	42	35	5	0.63	11.20	0.0027	Reject H_0	Strong evidence of improvement
100	42	34	4.5	0.45	17.78	0.0027	Reject H_0	Very strong evidence

Do not reject H_0 No significant difference found Reject H_0 Significant difference found Reject H_0 Strong/very strong evidence

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Sample Data For Three Hypothesis Testing Topics

Topic 1: Sleep & Memory | Topic 2: Fertilizer & Crop Yield | Topic 3: Headphones & Focus

H_0: Using noise-cancelling headphones does not significantly reduce task completion time.
H_1: Participants using noise-cancelling headphones complete tasks significantly faster.

Total participants / plots	Measurement unit	Test type	Significance level
32 (16 per group)	**task completion time in minutes**	**Two-tailed Z-test**	**α = 0.05**

Control group — no headphones

Participants worked in an open office environment with ambient noise and no headphones.

ID	Gender	Age	Time (min)
P01	Male	26	44
P02	Female	28	41
P03	Male	30	45
P04	Female	27	43
P05	Male	29	46
P06	Female	25	40
P07	Male	31	42
P08	Female	28	44
P09	Male	26	45
P10	Female	29	41
P11	Male	27	43
P12	Female	30	46
P13	Male	28	42
P14	Female	26	44
P15	Male	31	45
P16	Female	27	43
Mean			43.38
Std dev			1.76
n			16

Experiment group — noise-cancelling headphones

Participants used noise-cancelling headphones in the same open office while completing identical tasks.

ID	Gender	Age	Time (min)
P17	Male	26	37
P18	Female	28	35
P19	Male	30	38
P20	Female	27	36
P21	Male	29	34
P22	Female	25	36
P23	Male	31	35
P24	Female	28	37
P25	Male	26	34
P26	Female	29	36
P27	Male	27	35
P28	Female	30	33
P29	Male	28	36
P30	Female	26	35
P31	Male	31	34
P32	Female	27	36
Mean			35.44
Std dev			1.27
n			16

Each row is one participant / plot. Mean and SD are computed from the 16 data points shown and feed directly into the Z-test table.

Conclusion

In this experiment, hypothesis testing was performed using a two-tailed Z-test with a significance level of α = 0.05. Data from control and experiment groups was analyzed by calculating the mean, standard deviation, standard error, Z-statistic, and p-value. The p-values were compared with the significance level to accept or reject the null hypothesis. The experiment showed how hypothesis testing helps determine whether differences between groups are statistically significant or caused by random variation.